Immortal coils

***p. 30* … impossible to disentangle the contribution of one gene from that of another.**

Here, and also on pages 110–13, is my answer to critics of genetic 'atomism'. Strictly it is an anticipation, not an answer, since it predates the criticism! I am sorry that it will be necessary to quote myself so fully, but the relevant passages of *The Selfish Gene* seem to be disquietingly easy to miss! For example, in 'Caring Groups and Selfish Genes' (in *The Panda's Thumb*), S. J. Gould stated:

> There is no gene 'for' such unambiguous bits of morphology as your left kneecap or your fingernail. Bodies cannot be atomized into parts, each constructed by an individual gene. Hundreds of genes contribute to the building of most body parts …

Gould wrote that in a criticism of *The Selfish Gene*. But now look at my actual words (p. 30):

> The manufacture of a body is a cooperative venture of such intricacy that it is almost impossible to disentangle the contribution of one gene from that of another. A given gene will have many different effects on quite different parts of the body. A given part of the body will be influenced by many genes, and the effect of any one gene depends on interaction with many others.

And again (p. 46):

> However independent and free genes may be in their journey through the generations, they are very much *not* free and independent agents in their control of embryonic development. They collaborate and interact in inextricably complex ways, both with each other, and with their external environment. Expressions like 'gene for long legs' or 'gene for altruistic behaviour' are convenient figures of speech, but it is important to understand what they mean. There is no gene which single-handedly builds a leg, long or short. Building a leg is a multigene cooperative enterprise. Influences from the external environment too are indispensable; after all, legs are actually made of food! But there may well be a single gene which, *other things being equal*, tends to make legs longer than they would have been under the influence of the gene's allele.

I amplified the point in my next paragraph by an analogy with the effects of fertilizer on the growth of wheat. It is almost as though Gould was so sure, in advance, that I must be a naïve atomist, that he overlooked the extensive passages in which I made exactly the same interactionist point as he was later to insist upon.

Gould goes on:

> Dawkins will need another metaphor: genes caucusing, forming alliances, showing deference for a chance to join a pact, gauging probable environments.

In my rowing analogy (pp. 110–12), I had already done precisely what Gould later recommended. Look at this rowing passage also to see why Gould, though we agree over so much, is wrong to assert that natural selection 'accepts or rejects entire organisms because suites of parts, interacting in complex ways, confer advantages'. The true explanation of the 'cooperativeness' of genes is that: